

VIT BHOPAL UNIVERSITY

ACADEMIC PROJECT REPORT

V-SHOP E-COMMERCE SYSTEM

A Console-Based Shopping Application in Python

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Project Overview

The **V-SHOP E-Commerce System** is a functional console-based application designed using Python. Built as an academic milestone project, the application demonstrates core programming methodologies through a simulated e-commerce workflow, including secure registration via mock OTP validation, nested product catalogs, and automatic cash-on-delivery routing with dynamic delivery date computation.

Unlike massive web systems, this application operates locally without external service calls, databases, or graphical elements, highlighting foundational algorithmic logic and robust console user-experience practices.

Core Application Features

- **Authentication & Safety:** Verifies mobile contact structural formats (10 digits) and executes a randomized 4-digit OTP simulation to grant access.
- **Dynamic Cart & Catalog Flow:** Features deep conditional trees parsing major retail tiers: Clothes (Female/Male/Children with multi-tier size metrics), Hair Care, Body Care, and Footwear.
- **E-Commerce Logistics Module:** Auto-calculates complimentary shipping incentives, enforces Cash On Delivery protocols due to local sandboxing, and computes real-time future time deltas using standard time libraries.
- **Receipt Generation:** Performs real-time item aggregation, currency tallies, and clean timestamp printing mapped to local operational calendar indices.

Technologies Implemented

- **Python 3.x** Core Engine Core architecture build.
- **datetime & calendar** modules for calculating exact relative delivery windows.
- **random** library to compute secure validation codes and log delivery variables.
- **Control Structs:** Complex nested conditional architecture mapping hundreds of granular selections cleanly.

Project Architecture & File Layout

The program is structured cleanly inside an autonomous file layer designed to run directly inside typical modern terminal shells:

```
V_Shop_System/  
■  
■■■ main.py           # Main operational routing engine  
■■■ README.md        # Technical guidelines documentation
```

How to Execute the Program

Follow these standard steps to build and initialize the operational runtime environment:

Step 1: Setup Workspace – Initialize a clear path directory named V_Shop_System.

Step 2: Initialize Script – Save the structured python block into a file named main.py.

Step 3: Launch Shell Terminal – Direct your system shell platform into the folder root path.

Step 4: Execute Engine – Type and invoke: python main.py or python3 main.py.

Program Interface Specification

Upon initialization, the script spins up its visual prompt boundaries precisely like this:

```
.....WELCOME TO V-shop.....  
  
Enter your name= Vanshika  
Enter your phone no.= 9876543210  
OTP has been sent to your mobile no.=XXXXXX210  
                                otp=4821  
Enter otp= 4821
```

Program Execution Outputs (Simulated Visual Layouts)

The sections below illustrate precise interactive interface footprints generated directly by the program execution workflows.

Scenario A: Category Navigation & Cart Verification

```
Address:--
1.Your current location
2.Add new location
Your choice=1
Location = <<<<VIT BHOPAL UNIVERSITY>>>>
Hello, dear coustomer Vanshika
    Welcome to V-Shop

    !!!FIND WHAT YOU LOVE■,
        LOVE WHAT YOU FIND■!!!
Products:---
1.CLOTHES■
2.HAIR CARE■
3.BODYCARE■
4.FOOTWEAR■
Enter your preference:--1
1.Female
2.Male
3.Children
Enter your choice:--1
1.XS... 6.XXL
Enter your size :--3
1.Dresses | 2.Kurties | 3.Bottom Wear | 4.Tops & T-shirts
Enter your favorite wear:--1
1.Sage Meadow Dress in Green Botanical Print ~■3,299
2.Women's Orange Georgette Floral Printed Tiered Dress ~■800
Add the item to cart=1
Your cart value is :-- ■3,299
```

Scenario B: Logistics Processing & Dynamic Receipt Generation

```
With this cart you got free delivery■
press enter for proceed to pay
OOPS!!■
    Online payment mode is not avaiable!!
Only Cash On delivery is avaiable
Press enter to confirm order in COD mode
    Congrats ■ Your Order is placed
Delivery will be done by Saturday, 17 October 2026 at 03:42 PM

Want to shop again ?
Enter 1 or Yes to continue; anything else to exit: no
Your purchase(s):--
Date and time: Wednesday 2026-09-30 00:53:21

Sage Meadow Dress in Green Botanical Print ~■3,299

Total spend = 3299

.....Thank you for shopping from V-SHOP!.....
```